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Research and Science Services
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Education

- 2021 **Ph.D. in Astronomy**
 University of Surrey, Guildford, UK
Thesis: Stellar history of the SMC in the era of deep photometric surveys
Supervisor: Noelia E. D. Noël
- 2017 **Masters in Astrophysics and Cosmology**
 Universitat de Barcelona, Barcelona, Catalonia
Thesis: Scientific Validation of Clusterix 2.0
Supervisor: Carme Jordi
- 2016 **Degree in Physics (BSc equivalent)**
 Universitat de Barcelona, Barcelona, Catalonia

Research Appointments

- 2024–now Research Fellow, NSF's NOIRLab, La Serena, Chile
- 2022–2024 Postdoctoral Researcher, Montana State University, United States
- 2020–2021 Support Astronomer at Isaac Newton Group of Telescopes, La Palma, Spain
- 2015–2017 Undergraduate Research Associate doing Light Pollution studies, Universitat de Barcelona

Teaching and Advising

- 2025–2026 Mentor for 3 students of REU and PIA undergraduate programs at NOIRLab Chile.
- 2024 Teaching “Stars and the Milky Way” (Spring semester) at Montana State University.
- 2022–now External co-supervisor for Mahdiah Navabi (PhD student), University of Surrey, Guildford
- 2022–2024 Teaching “Introduction to Python”, tailored to astronomy students.
- 2018–2020 Advising Kasia Nowak, Alex Goater and Aoife Simpson (Masters Physics students) part of a Summer placement. Two of them won prizes for posters made based on the project they developed.
- Teaching Assistant for “Scientific Investigation Skills”, “Introduction to Astronomy”, and “Physics First year Undergraduate Laboratories” modules, University of Surrey, Guildford

Publications

- 2026 M. Gatti, and 39 co-authors, including **P. Massana**, *DECADE+DES Y3 weak lensing mass map: A 13,000 deg² view of cosmic structure from 270 million galaxies*, PhRvD, 113, L081303.
- C. Y. Tan, and 88 co-authors, including **P. Massana**, *DELVE Milky Way Satellite Galaxy Census. I. Satellite Population and Survey Selection Function in DES, DELVE, and Pan-STARRS*, ApJ, 1000, 87.
- 2025 J. T. Povick, and 11 co-authors, including **P. Massana**, *Revealing the chemical structure of the Magellanic Clouds with APOGEE. III. Abundance gradients of the Small Magellanic Cloud*, MNRAS, 544, 430.

- D. Anbajagane, and 52 co-authors, including **P. Massana**, *The DECADE cosmic shear project III: validation of analysis pipeline using spatially inhomogeneous data*, OJAp, 8, 46160.
- D. Anbajagane, and 52 co-authors, including **P. Massana**, *The DECADE cosmic shear project II: photometric redshift calibration of the source galaxy sample*, OJAp, 8, 46159.
- D. Anbajagane, and 53 co-authors, including **P. Massana**, *The DECADE cosmic shear project I: A new weak lensing shape catalog of 107 million galaxies*, OJAp, 8, 46158.
- P. L. Schechter, and 27 co-authors, including **P. Massana**, *The DELVE Quadruple Quasar Search. I. A Lensed Low-luminosity Active Galactic Nucleus*, AJ, 170, 241.
- V. M. Placco, and 25 co-authors, including **P. Massana**, *The DECam MAGIC Survey: Spectroscopic Follow-up of the Most Metal-poor Stars in the Distant Milky Way Halo*, ApJ, 991, 101.
- A. B. Pace, and 32 co-authors, including **P. Massana**, *Spectroscopic Analysis of Pictor II: a very low metallicity ultra-faint dwarf galaxy bound to the Large Magellanic Cloud*, OJAp, 8, 112.
- A. Viswanathan, and 29 co-authors, including **P. Massana**, *The Pristine survey: XXV. The very metal-poor Galaxy: Chemodynamics through the follow-up of the Pristine-Gaia synthetic catalogue*, A&A, 695, A112.
- W. Cerny, and 39 co-authors, including **P. Massana**, *Discovery and Spectroscopic Confirmation of Aquarius III: A Low-mass Milky Way Satellite Galaxy*, ApJ, 979, 164.
- 2024 J. D. Sakowska, and 14 co-authors, including **P. Massana**, *Unveiling the purely young star formation history of the SMC's northeastern shell from colour-magnitude diagram fitting*, MNRAS, 532, 4272. (contributed science)
- M. De Leo, J. I. Read, N. E.D. Noël, and 3 more coauthors, including **P. Massana**, *Surviving the Waves: evidence for a Dark Matter cusp in the tidally disrupting Small Magellanic Cloud*, MNRAS, 535, 1015. (contributed science)
- J. Povick, D. Nidever, **P. Massana**, and 12 more coauthors, *Revealing the Chemical Structure of the Magellanic Clouds with APOGEE. I. Calculating Individual Stellar Ages of RGB Stars in the Large Magellanic Cloud*, MNRAS, 533, 3685. (contributed science)
- P. Massana**, D. Nidever, K. Olsen, *The Magellanic Puzzle: origin of the periphery*, MNRAS, 527, 8706.
- 2023 W. Cerny, A. Drlica-Wagner, T. Li and 24 more coauthors, including **P. Massana**, *DELVE 6: An Ancient, Ultra-faint Star Cluster on the Outskirts of the Magellanic Clouds*, ApJL, 953, L21.
- W. Cerny, C. Martínez-Vázquez, A. Drlica-Wagner and 31 more coauthors, including **P. Massana**, *Six More Ultra-faint Milky Way Companions Discovered in the DECam Local Volume Exploration Survey*, ApJ, 953, 1.
- 2022 C. Bell, M-R. L. Cioni, A. Wright and 15 more coauthors, including **P. Massana**, *The intrinsic reddening of the Magellanic Clouds as traced by background galaxies - III. The Large Magellanic Cloud*, MNRAS, 516, 824.
- A. Drlica-Wagner, P. Ferguson, M. Adamów and 125 more coauthors, including **P. Massana**, *The DECam Local Volume Exploration Survey Data Release 2*, ApJS, 261, 38.
- P. Massana**, T. Ruiz-Lara, Carme Gallart, N.E.D. Noël, and SMASH Collaboration, *The synchronised dance of the Magellanic Clouds' star formation history*, MNRAS, 513, L40.
- 2021 W. Cerny, A. B. Pace, A. Drlica-Wagner, and 29 more coauthors, including **P. Massana**, *Discovery of an Ultra-Faint Stellar System near the Magellanic Clouds with the DECam Local Volume Exploration (DELVE) Survey*, The Astrophysical Journal, 910, 18.
- D.L. Nidever; K. Olsen; Y. Choi and 32 more coauthors, including **P. Massana**, *The Second*

Data Release of the Survey of the Magellanic Stellar History (SMASH), The Astronomical Journal, 161, 74.

- 2020 C. P. M. Bell, M-R. L. Cioni, A. H. Wright, and 23 more coauthors, including **P. Massana**, *The intrinsic reddening of the Magellanic Clouds as traced by background galaxies - II. The Small Magellanic Cloud*, Monthly Notices of the Royal Astronomical Society, 499, 993.

P. Massana, N.E.D. Noël, D.L. Nidever, and SMASH Collaboration, *SMASHing the low surface brightness SMC*, 2020, MNRAS, 498, 1034.

T. Ruiz-Lara, C. Gallart, M. Monelli, and more 23 coauthors, including **P. Massana**, *The Large Magellanic Cloud stellar content with SMASH. I. Assessing the stability of the Magellanic spiral arms*, Astronomy & Astrophysics, 639, L3. (contributed science)

S. Mau, W. Cerny, A. B. Pace, and 56 more coauthors, including **P. Massana**, *Two Ultra-faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey*, The Astrophysical Journal, 890, 136.

- 2019 D.L. Nidever, K. Olsen, Y. Choi, and more 23 coauthors, including **P. Massana**, *Exploring the Very Extended Low-surface-brightness Stellar Populations of the Large Magellanic Cloud with SMASH*, 2019, The Astrophysical Journal, 874, 118.

- 2018 Y. Choi, D.L. Nidever, K. Olsen, and more 18 coauthors, including **P. Massana**, *SMASHing the LMC: Mapping a Ring-like Stellar Overdensity in the LMC Disk*, 2018, The Astrophysical Journal, 869, 125.

Y. Choi, D.L. Nidever, K. Olsen, and more 18 coauthors, including **P. Massana**, *SMASHing the LMC: A Tidally Induced Warp in the Outer LMC and a Large-scale Reddening Map*, 2018, The Astrophysical Journal, 866, 90.

Presentations (as first author only)

- 2025 *New horizons for the MC periphery*, XMC II Clouds over Yellowstone, Bozeman, MT, USA. Given 28/05/25.

DELVEing into the Magellanic Clouds: a complete view, AAS Winter meeting 2025, National Harbor, MD, USA. Given 14/01/25.

- 2024 *The variable sky with the NOIRLab Source Catalog*, Rare Gems in Big Data, Tucson, AZ, USA. Given 20/05/24.

- 2023 *The Magellanic Puzzle: origin of the periphery*, RelAstro seminars, Bozeman, MT, USA. Given 20/09/23.

The synchronized dance of the Magellanic Clouds' star formation history, AAS Winter meeting 2023, Seattle. Given 12/01/23.

- 2022 *Unravelling the history of the Magellanic Clouds with SMASH*, DECam at 10 years, Tucson. Given 12/09/22.

- 2021 *Unravelling the history of the Magellanic Clouds with SMASH*, ICCUB seminar, University of Barcelona. Given 20/01/21.

- 2020 *Star Formation History of the SMC through SMASH*, contributed talk, Space Telescope Science Institute Spring Symposium, The Local Group: Assembly and Evolution, Baltimore, US (virtual). Given 02/09/20.

Star Formation History of the SMC through SMASH, contributed talk, European Astronomical Society Annual meeting, Leiden, Netherlands (virtual). Given 01/07/20.

SMASHing the Small Magellanic Cloud (review of my PhD until date), internal seminar, University of Surrey, UK. Given 11/05/20. Recorded virtually and available here <https://youtu.be/eIG4FFSVHNw>.

- 2019 *About the diet of galaxies*, invited talk, Institute of Physics lectures, University of Surrey, UK. Given 12/11/19.

SMASHing The Small Magellanic Cloud, contributed poster and flash talk, Small Galaxies, Cosmic Questions, Durham, UK. 29/7/19–2/8/19.

SMASHing the SMC: Investigating its profile, contributed talk, Student-led conference: From Infinity to Zero: A History of the Universe in Redshift, University of Southampton, Southampton, UK. 3/4/19–5/4/19.

Service & Outreach

2025–2027	REU/PIA programs, Site coordinator
2023–2026	DELVE Survey¹, MC group coordinator
2022–	Journal Club organiser for Astrophysics group at Montana State University
2020–2021	SMASH monthly meetings organiser
2017–2020	Journal Club organiser for Surrey Astrophysics Group
	Applicant and Open day student helper
	Volunteer, Surrey Stargazing Public Events
2018	Surrey Stargazing speaker

¹<https://sites.google.com/view/delve-survey>